NYXORA — National Hackathon 2026

ZeroWaste

AI-Powered Surplus Food Redistribution Platform

Connecting restaurants, NGOs, and volunteers in real-time to rescue surplus food before it goes to waste.

Donors

Restaurants & Hotels

NGOs

Food Banks & Trusts

Volunteers

Last-mile Delivery

TEAM ERROR 404

Team & Problem Statement

TEAM NAME

Error 404

Not Found — until now.

PROJECT TITLE

ZeroWaste

AI-Powered Surplus Food Redistribution Platform

PROBLEM STATEMENT

Every day, **40% of food produced** in India is wasted — mostly by restaurants, hotels, and catering businesses — while **189 million people** go to sleep hungry.

There is no efficient, accountable, real-time system to bridge this gap. Existing solutions rely on phone calls, manual coordination, and have no delivery accountability.

The result: edible food rots in kitchens while people starve within kilometers.

Our Solution

ZeroWaste is a multi-role real-time platform that digitises the full food donation lifecycle — from posting surplus, to claiming, to OTP-verified delivery.

Donor Portal

Restaurants & hotels post surplus food with geo-location, expiry time, images, and food details. Auto-fill address from profile or live GPS.

NGO Dashboard

NGOs browse and instantly claim listings — full or partial. Real-time tracker shows volunteer status, ETA, and auto-generates delivery OTP.

Volunteer App

Volunteers see tasks sorted by distance. Accept, view route map, verify pickup via donor OTP, confirm delivery via NGO OTP — earn automatically.

Key Innovation: Dual OTP verification at pickup and delivery creates a tamper-proof chain of custody. Wallet payouts are automatic — no manual processing.

Competitive Edge & USP

Zomato / Swiggy	Purely commercial — zero food donation mechanism
Manual Drives (Robin Hood Army)	Phone/WhatsApp based. No tracking, no accountability
Food Banks	Static drop-off model. No last-mile delivery
ZeroWaste	End-to-end digital. Auto-assigns volunteer. OTP-verified. Real-time for all 3 parties.

OUR USP

- **Dual OTP Verification**
- **Real-time Live Tracker**
- **Auto Wallet Payout**
- **AI Demand Forecast**

One-click posting → Auto proximity match → OTP chain prevents fraud → Financial incentive keeps supply healthy

Features

Geo-Location Posting

Live GPS or registered address. Distance shown to volunteers.

Auto Volunteer Assign

Nearest available volunteer auto-assigned on claim.

Dual OTP Verification

Pickup OTP → donor. Delivery OTP → NGO. Tamper-proof.

Real-time Tracker

Socket.IO live updates: assigned → picked up → delivered.

Wallet System

NGO wallet auto-debits, volunteer credited on delivery.

Route Map + ETA

OpenStreetMap with pickup/drop pins. Haversine ETA.

AI Demand Forecast

Groq LLaMA 3.3 predicts surplus days and peak times.

Real-time Notifications

Socket events + in-app bell. OTP popup auto-appears.

Role-based Dashboards

Separate UX for Donor, NGO, Volunteer, and Admin.

Process Flow

1. **Donor Posts Food** — + location, expiry
2. **NGO Claims** — real-time browse
3. **Volunteer Assigned** — nearest available
4. **Pickup OTP Verified** — donor shows code
5. **En Route to NGO** — live tracking
6. **Delivery OTP Verified** — NGO shows code
7. **Auto Wallet Settlement** — instant payout

DONOR	NGO	VOLUNTEER
Register & set location	Browse & claim listings	Browse tasks by distance
Post food listing	Track volunteer live	Accept & view route
Show Pickup OTP	Show Delivery OTP	Enter pickup & delivery OTPs
View delivery history	Manage wallet	Receive payout

Key Screen Flows

Donor — Post Food	NGO — Live Tracker	Volunteer — OTP	Route Map	Earnings Wallet
Food Type: Cooked	⚡ LIVE ETA ~8 min	CONFIRM DELIVERY	🗺 OpenStreetMap	₹340 Total Earned
Items: Biryani 10kg, Dal 5L	✓ Claimed · 12:05	Ask NGO for their OTP	Pickup pin → Drop-off pin	5 Deliveries Completed
Expiry: 3 hours	🛵 Volunteer assigned · 12:06	[8] [4] [7] [2] [9] [1]	Distance: 2.1 km	+₹154 Fried Rice · today
Pickup: Hotel Saraswathi	📦 Picked up · 12:07		ETA: ~4 min	+₹86 Idli batch · yesterday
[Submit Listing]	OTP: 847291	[Confirm Delivery]		[Withdraw]

Live prototype running with 16 pre-seeded accounts — full end-to-end flow

System Architecture

CLIENT LAYER

Browser (4 roles) • React Components • Socket.IO Client

APP LAYER

Next.js 16 App Router • API Routes • NextAuth.js (JWT)

REALTIME LAYER

Socket.IO Server • Room-based events • Per-user targeting

DATA LAYER

MongoDB Atlas • Mongoose ODM • 2dsphere Geo Index

EXTERNAL APIs

Groq LLaMA 3.3 • Cloudinary • Nominatim • OpenStreetMap

KEY DATA MODELS

U — **User**
role, location (GeoJSON), wallet, phone

F — **FoodListing**
status, foodItems[], donor, OTP refs

D — **FoodDemand**
meals, urgency, NGO pull-model

O — **OTP**
6-digit, type, TTL, attempts lock

W — **WalletTxn**
amount, type, balanceAfter

T — **VolunteerTask**
status, distance, payout

N — **Notification**
userId, type, 30-day TTL

Technologies Used

FRONTEND
- Next.js 16 (App Router)
- React 18 + TypeScript
- TailwindCSS 4
- Framer Motion
- Recharts

BACKEND / API
- Next.js API Routes
- NextAuth.js (JWT)
- Zod Validation
- Custom server.ts

REAL-TIME
- Socket.IO 4
- Room-based routing
- Per-user targeting

DATABASE
- MongoDB Atlas
- Mongoose 9 ODM
- 2dsphere Geo Index
- TTL Indexes

AI / ML
- Groq SDK
- LLaMA 3.3 70B
- Statistical forecasting

MAPS & GEO
- React-Leaflet
- OpenStreetMap tiles
- Nominatim geocoding
- Haversine distance

MEDIA
- Cloudinary uploads
- Image optimization

SECURITY
- bcryptjs hashing
- OTP (6-digit, 24h TTL)
- Role-based middleware
- Audit logging

Deployed on Render (Singapore) • pnpm • Custom Node.js server for Socket.IO

Implementation Cost

MVP PHASE (Current — Free)

Component	Service	Cost/Mo
Hosting	Render (Free)	₹0
Database	MongoDB Atlas (M0)	₹0
Maps	OpenStreetMap + Nominatim	₹0
Auth	NextAuth (self-hosted)	₹0
AI	Groq (Free tier)	₹0
Media	Cloudinary (Free)	₹0
MVP Total		**₹0 / month**

PRODUCTION SCALE PHASE

Component	Service	Cost/Mo
Hosting	Render Pro	~₹1,700
Database	MongoDB M10	~₹2,100
SMS OTP	Twilio / MSG91	~₹500
CDN/Storage	Cloudflare R2	~₹400
Domain	.in domain	~₹100
AI	Groq Pro	~₹500
Production Total		**~₹5,300 / month**

Self-sustaining model: NGO wallet top-ups and a small platform fee (₹10/delivery) on high-volume corridors can cover operational costs without external funding.

Impact & Roadmap

40%

Food waste reduction
in partner areas

1000+

Meals redirected
per month per city

₹0

Cost to donors
just one click

ROADMAP

1. **Mobile App (React Native)**
   Native Android/iOS with push notifications and live GPS
2. **SMS OTP (Twilio / MSG91)**
   Works even without internet on the donor/NGO side
3. **Corporate CSR Integration**
   Tax-benefit receipts and bulk listing API for hotel ERPs
4. **Multi-city Expansion**
   City-level admin dashboards with impact metrics and leaderboards
5. **Government Integration**
   Connect with FSSAI food banks and state hunger-map data

ADDITIONAL CAPABILITIES

- **Pull Model (Demands)**
  NGOs post what they need → donors accept → volunteer delivers
- **Partial Claiming**
  Multiple NGOs claim different portions of the same listing
- **Admin Panel**
  User management, listing control, audit trail, analytics
- **Cron Expiry**
  Hourly job marks expired listings and notifies all parties
- **Rating System**
  Volunteers rated 0-5 stars, prioritized in auto-assignment

TEAM ERROR 404

ZeroWaste

Turning Surplus into Service

Because no plate of food should go to waste while someone goes to bed hungry.

GitHub Repository
https://github.com/addy9691/zerowaste.git

Live MVP Prototype
WEBSITE LINK
16 pre-seeded accounts • Full end-to-end flow